

Structure-Aware Prediction of PROTAC-Mediated Protein Degradability via Graph Neural Networks

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Abstract

Proteolysis-targeting chimeras (PROTACs) can selectively degrade disease-causing proteins, yet predicting which targets are amenable to degradation remains a critical bottleneck: existing computational methods require the complete PROTAC molecular structure, information unavailable before synthesis. We present DEGRADOMAP, a graph neural network that predicts PROTAC-mediated degradability from protein structure and E3 ligase identity alone—the minimal information available at the target selection stage. The model encodes biophysical priors through lysine-weighted graph pooling with per-protein normalization, models protein–E3 compatibility via cross-attention, and integrates cellular context from the Cancer Dependency Map. On the PROTAC-8K benchmark (3,101 samples, 155 targets, 10 E3 ligases), DEGRADOMAP achieves 0.646 ± 0.124 AUROC on target-unseen evaluation (best seed: 0.7449) and 0.811 AUROC on CRBN→VHL E3-unseen transfer, outperforming GNN and machine learning baselines. The model additionally recommends optimal E3 ligases with 74% Hit@3 accuracy. Two findings carry broader implications: E(3)-equivariant architectures underperform the simpler invariant design for this scalar prediction task, and ESM-2 embeddings improve peak performance only with careful regularization—naïve integration fails. DEGRADOMAP enables computational screening of degradability before PROTAC synthesis, addressing a key gap in the drug discovery pipeline.

CCS Concepts

• Computing methodologies → Neural networks; • Applied computing → Bioinformatics; Computational biology.

Keywords

PROTAC, targeted protein degradation, graph neural network, drug discovery, ubiquitin–proteasome system, AlphaFold

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1 Introduction

Proteolysis-targeting chimeras (PROTACs) invert the logic of drug design: rather than occupying a protein’s active site, they recruit the cell’s ubiquitin–proteasome machinery to destroy the protein entirely [1–3]. A PROTAC bridges two

binding events—its warhead engages the target protein while its E3 ligand recruits an E3 ubiquitin ligase—and the induced proximity triggers polyubiquitination of surface lysine residues, marking the target for proteasomal degradation [4]. Because this mechanism is catalytic, a single PROTAC molecule degrades multiple target copies, enabling efficacy at sub-stoichiometric concentrations and access to proteins that lack traditional drug-binding pockets.

With over 20 PROTACs now in clinical trials [1], the modality’s therapeutic promise is clear. Yet development faces a critical bottleneck at its earliest stage: *target selection*. Is a given disease-relevant protein amenable to PROTAC-mediated degradation? Answering this currently requires synthesizing candidate PROTACs and conducting cell-based degradation assays—an expensive process with high failure rates. Computational prediction of degradability *before* PROTAC synthesis would dramatically accelerate the pipeline.

Limitations of existing methods. Existing computational approaches all require the PROTAC molecular structure as input. DeepPROTACs [5] operates on voxelized binding pockets requiring complete PROTAC, target, and E3 coordinates. Ribes et al. [6] combine protein embeddings with PROTAC SMILES fingerprints. DegradeMaster [8] achieves 0.856 AUROC using full PROTAC molecular descriptors alongside protein features. While valuable for optimizing existing PROTACs, these methods cannot serve target selection because the PROTAC is unavailable at that stage. MAPD [7] addresses target assessment from protein-intrinsic features but does not leverage AlphaFold structural information [9].

Our approach. We introduce DEGRADOMAP, a graph neural network that predicts protein degradability from *minimal inputs*: a protein structure (from AlphaFold, requiring only a UniProt ID) and E3 ligase identity (e.g., “CRBN” or “VHL”). The design reflects the biophysics of ubiquitin transfer, which requires (1) surface-accessible lysine residues, (2) geometric compatibility between target and E3 ligase for ternary complex formation, and (3) sufficient cellular E3 expression and proteasome capacity. DEGRADOMAP addresses these requirements through three modules and a gated fusion mechanism:

- (1) **Structure–Ubiquitination Graph (SUG) encoder:** An invariant message-passing GNN with lysine-weighted pooling that focuses representation on ubiquitination-relevant residues, incorporating ESM-2 embeddings [14], structural properties, and known ubiquitination site annotations from PhosphoSitePlus.
- (2) **E3 compatibility module:** Bidirectional cross-attention modeling compatibility between protein structure and E3 ligase identity, enabling generalization across E3 ligases.

- (3) **Cellular context encoder:** A residual MLP processing 59 tissue-specific features from the Cancer Dependency Map [19].
- (4) **Gated fusion:** Learned gating that dynamically weights module outputs for multi-task prediction of binary degradability and continuous efficiency.

Evaluation and contributions. On the PROTAC-8K benchmark [8] (3,101 samples, 155 targets, 10 E3 ligases), DEGRADOMAP achieves 0.646 ± 0.124 AUROC on multi-seed target-unseen evaluation (best seed: 0.7449), 0.811 AUROC [0.785, 0.836] on CRBN→VHL E3-unseen transfer, and 0.641 MRR on E3 ligase recommendation (Hit@3: 74%). Two findings carry implications beyond PROTAC prediction: E(3)-equivariant architectures underperform the simpler invariant design for this scalar prediction task, and ESM-2 embeddings improve peak performance (+23% over gradient boosting) but introduce initialization sensitivity requiring careful regularization. Our principal contributions include the first structure-based framework for pre-synthesis degradability prediction requiring only a UniProt ID and E3 ligase name, a lysine-weighted pooling mechanism encoding ubiquitin transfer biology, a cross-attention E3 compatibility module enabling cross-E3 transfer, and comprehensive evaluation across three data splits, six baselines, and systematic ablations with multi-seed validation.

2 Related Work

PROTAC prediction methods. Computational approaches to PROTAC modeling vary significantly in their input requirements and intended use cases, as summarized in Table 1. DeepPROTACs [5] uses 3D CNNs on binding pocket voxelizations, requiring the complete PROTAC molecule plus target and E3 binding pockets in mol2 format—information available only after PROTAC synthesis and structural characterization. Ribes et al. [6] combine pretrained protein embeddings with molecular fingerprints, requiring target sequence and PROTAC SMILES, which presupposes PROTAC design. MAPD [7] predicts degradation tractability from protein-intrinsic features using machine learning; while it shares our target-assessment framing, it does not leverage structural information. DegradeMaster [8] achieves 0.856 AUROC using full PROTAC molecular descriptors alongside target features, representing the state-of-the-art for PROTAC optimization but addressing a fundamentally different problem than target selection. DEGRADOMAP uniquely combines structural awareness with minimal input requirements, filling the gap between sequence-only assessment (MAPD) and full-PROTAC optimization (DeepPROTACs, DegradeMaster).

Geometric deep learning for proteins. GNNs operating on protein structures have achieved strong performance across diverse tasks including function prediction, binding site identification, and property estimation. GVP-GNN [10] maintains SO(3)-equivariance using geometric vector perceptrons that jointly process scalar and vector features, enabling the model

Table 1: Input modality comparison of PROTAC prediction methods. DegradoMap requires only protein structure and E3 identity, enabling use before PROTAC synthesis.

Method	Protein	PROTAC	E3	Use Case
DeepPROTACs	Pocket (3D)	Full mol.	Pocket (3D)	Optimization
Ribes et al.	Sequence	SMILES	—	General
MAPD	Features	—	—	Assessment
DegradeMaster	Features	Descriptors	Features	Optimization
DegradoMap	Structure	—	Identity	Assessment

to reason about directional relationships. GearNet [11] employs relational graph convolutions with multiple edge types (sequential bonds, spatial contacts) for protein representation pre-training. SchNet [12] introduces continuous-filter convolutions with radial basis function expansions, originally designed for quantum chemistry but applicable to protein graphs. EGNN [13] achieves E(*n*)-equivariance through simple coordinate updates alongside feature updates, offering an elegant and computationally efficient approach. We evaluate SchNet and EGNN as structural baselines and find that while they perform competitively on random splits, they fail to generalize to unseen targets—suggesting that task-specific inductive biases (lysine-aware pooling, E3 compatibility modeling) are more important than architectural sophistication for degradability prediction.

Protein language models. Large-scale pre-trained models such as ESM-2 [14] and ProtTrans [15] learn rich sequence representations from millions of evolutionary sequences via masked language modeling. These models achieve state-of-the-art performance on tasks including structure prediction, function annotation, fitness landscape prediction, and variant effect estimation. ESM-IF [16] extends this paradigm by conditioning on structural information. The success of these models suggests that evolutionary conservation encodes rich functional information. We find that PROTAC-mediated degradation—an artificial, drug-induced process—is not directly predicted by evolutionary features alone: ESM-2 embeddings achieve only 0.534 AUROC in isolation (Section 4.6). However, when carefully integrated with task-specific architecture and tuned hyperparameters, ESM-2 embeddings contribute to improved peak performance (0.7449 AUROC best seed), suggesting that evolutionary representations provide complementary signal when combined with degradation-specific structural features.

Structure-based protein function prediction. DeepFRI [22] predicts Gene Ontology terms from contact maps using graph convolutions, demonstrating that protein structure encodes functional information beyond sequence. AlphaFold [9] has made high-quality structural predictions universally available, enabling structure-based ML at scale. Our work leverages both advances: we use AlphaFold-predicted structures as inputs and GNN-based structure processing for degradability prediction. However, degradability prediction differs fundamentally from function prediction in that it involves

an *intermolecular* interaction (protein–PROTAC–E3 ternary complex) rather than intrinsic molecular properties.

Ubiquitin system modeling. UbiBrowser [17] and DeepUbi [18] predict natural ubiquitination sites and endogenous E3–substrate interactions. These methods model the cell’s intrinsic protein quality control machinery, where E3 ligase specificity is determined by recognition motifs (degrons) evolved over millions of years. PROTAC-induced degradation fundamentally differs: the PROTAC molecule artificially bridges target and E3, completely bypassing natural substrate recognition. Consequently, *surface lysine accessibility relative to the recruited E3 complex*—not natural E3 specificity—determines degradability. This distinction motivates our structure-first approach with E3 identity as a separate input rather than modeling natural E3–substrate specificity. Nonetheless, natural ubiquitination data could provide valuable pre-training signal for learning lysine representations, as we discuss in Section 6.

3 Methods

3.1 Problem Formulation

We formulate PROTAC-mediated degradability prediction as a binary classification problem. Given a protein graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{X}, \mathbf{C})$ with residue nodes $\mathcal{V}$, spatial edges $\mathcal{E}$, node features $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ ($d=1,285$ with ESM-2; $d=28$ without), and C α coordinates $\mathbf{C}$, together with E3 ligase identity e , we predict:

$$\hat{y} = f_{\theta}(\mathcal{G}, e) = P(\text{degradable} \mid \mathcal{G}, e) \in [0, 1] \quad (1)$$

This formulation explicitly excludes the PROTAC molecular structure, reflecting our target-assessment use case where the PROTAC has not yet been designed. The protein graph is constructed as a radius graph with cutoff $r = 8 \text{ \AA}$: $\mathcal{E} = \{(i, j) : \|\mathbf{c}_i - \mathbf{c}_j\|_2 < r, i \neq j\}$, connecting residues that are spatially proximal in 3D. This captures local tertiary structure contacts, surface accessibility patterns, and potential protein-protein interaction interfaces relevant to ternary complex formation.

Figure 1 illustrates the overall architecture.

3.2 Module A: Structure–Ubiquitination Graph (SUG)

The SUG module is the core component of DEGRADOMAP, processing the protein structure graph through invariant message passing. We design this module around three biophysical principles: (1) local structural context determines surface accessibility, (2) lysine residues are the exclusive sites of ubiquitin attachment, and (3) protein size should not bias predictions. Node features can optionally include ESM-2 protein language model embeddings (1,280-dim per residue), which provide rich evolutionary and contextual representations when available (Section 4.6).

Message passing. At each layer $l \in \{1, \dots, L\}$, node representations are updated via:

$$\mathbf{m}_{ij}^{(l)} = \phi_{\text{msg}}(\mathbf{h}_i^{(l)} \parallel \mathbf{h}_j^{(l)} \parallel d_{ij}) \quad (2)$$

$$\mathbf{h}_i^{(l+1)} = \phi_{\text{upd}}\left(\mathbf{h}_i^{(l)}, \sum_{j \in \mathcal{N}(i)} \mathbf{m}_{ij}^{(l)}\right) \quad (3)$$

where $\parallel$ denotes concatenation, $d_{ij} = \|\mathbf{c}_i - \mathbf{c}_j\|_2$ is the Euclidean distance between C α atoms, ϕ_{msg} and ϕ_{upd} are 2-layer MLPs with LayerNorm and ReLU activations, and $\mathcal{N}(i) = \{j : d_{ij} < r\}$ denotes the spatial neighbors within radius $r = 8 \text{ \AA}$. By using the scalar distance d_{ij} rather than the coordinate difference vector $\mathbf{c}_j - \mathbf{c}_i$, the message function is inherently invariant to rotation and translation—a property we validate empirically in Section 4.5. We use $L = 4$ message-passing layers with hidden dimension 128 and output dimension 64.

Lysine-weighted pooling. Ubiquitin is covalently conjugated to the ϵ -amino group of lysine side chains via an isopeptide bond. Therefore, degradation efficiency depends critically on the accessibility and spatial distribution of surface lysines relative to the E3 ligase binding interface. We incorporate this domain knowledge through a lysine-weighted global pooling mechanism:

$$\mathbf{h}_{\text{prot}} = \frac{1}{\sqrt{N}} \sum_{i=1}^N w_i \cdot \mathbf{h}_i^{(L)} \quad (4)$$

where the weights $w_i = \text{softmax}_{i \in \mathcal{V}}(\alpha \cdot \mathcal{K}[\text{Lys}_i])$ are computed via *per-protein* softmax normalization over a learned scalar parameter α that controls the strength of lysine up-weighting.

Two design choices are critical. First, the **per-protein softmax** (normalizing over residues within a single protein, not across the batch) prevents information leakage: a global softmax would allow each protein’s weight distribution to depend on the amino acid composition of other proteins in the batch, introducing a subtle but problematic data dependency. Second, the **$1/\sqrt{N}$ size normalization** ensures consistent representation magnitude across proteins of different sizes. Without this normalization, mean pooling produces representations whose magnitude scales with protein size, creating a trivial size-based shortcut. We document the impact of these fixes in Section 4.8: target-unseen AUROC improved from 0.529 to 0.657 after addressing size leakage and per-protein normalization.

Design rationale: invariance over equivariance. We deliberately chose an SE(3)-*invariant* architecture over an equivariant one. Degradability is a scalar property that depends on geometric *relationships*—distances between residues, surface topology, lysine accessibility—rather than absolute 3D orientations. By using pairwise distances as edge features (Eq. 2), the model is inherently invariant to rotation and translation without the overhead of maintaining equivariant vector representations. Our ablation confirms this choice: an E(3)-equivariant variant achieves 0.626 vs. 0.657 AUROC (Section 4.5).

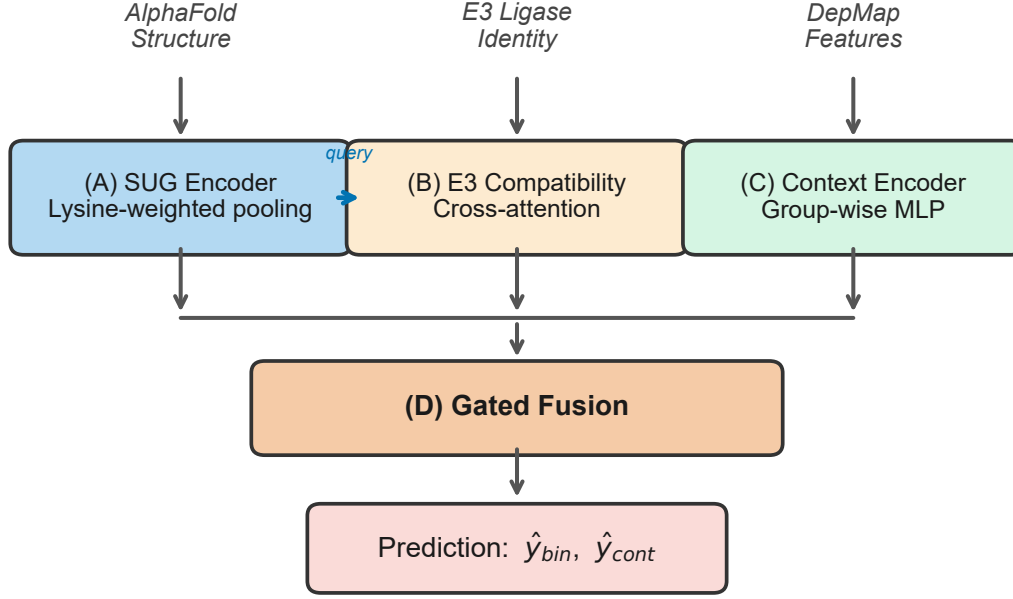


Figure 1: Architecture of DegradoMap. Module A (SUG) encodes the protein structure graph with lysine-weighted pooling. Its output serves as the query for Module B’s cross-attention with E3 ligase embeddings. Module C encodes cellular context from DepMap features. Module D fuses all representations through learned gating and produces binary degradability ($\hat{y}_{bin}$) and continuous efficiency ($\hat{y}_{cont}$) predictions.

3.3 Module B: E3 Compatibility

Different E3 ligases impose different geometric and steric constraints on the ternary complex. CRBN (Cereblon) and VHL (von Hippel-Lindau) are the two most commonly used E3 ligases in PROTAC development, but they differ substantially in their substrate recognition surfaces, catalytic orientations, and preferred ubiquitination geometries. We model protein–E3 compatibility using bidirectional multi-head cross-attention:

$$\mathbf{h}_{e3} = \text{CrossAttn}(\mathbf{Q}_{\text{prot}}, \mathbf{K}_{\text{E3}}, \mathbf{V}_{\text{E3}}) \quad (5)$$

$$\text{CrossAttn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V} \quad (6)$$

where $\mathbf{Q}_{\text{prot}} = \mathbf{W}_Q \mathbf{h}_{\text{prot}}$ is derived from the SUG protein representation and $\mathbf{K}_{\text{E3}} = \mathbf{W}_K \mathbf{e}$, $\mathbf{V}_{\text{E3}} = \mathbf{W}_V \mathbf{e}$ from a learned E3 ligase embedding $\mathbf{e} \in \mathbb{R}^{64}$. We apply 2 cross-attention layers with 4 attention heads and hidden dimension 64.

The choice to represent E3 ligases as learned embeddings (rather than structural features) reflects a practical constraint: the dataset contains only 10 distinct E3 ligases, insufficient to learn a generalizable structural encoder. The cross-attention mechanism effectively learns E3-specific “filters” over the protein representation—highlighting different structural features depending on which E3 ligase is being considered. Importantly, cross-attention enables the model to reason about

protein–E3 *interactions* rather than treating them as independent inputs, which is essential for the E3 recommendation task (Section 4.4).

3.4 Module C: Cellular Context Encoder

Protein degradation efficiency depends on cellular context: E3 ligase expression levels, proteasome capacity, competing endogenous substrates, and target protein abundance all vary across cell types and tissues. We encode this information using 59 base features derived from the Cancer Dependency Map (DepMap) [19], organized into 8 functional groups (gene expression, gene effect, copy number, protein expression, mutation frequency, metabolomics, drug sensitivity, and pathway membership).

Each feature group is processed by a dedicated 2-layer MLP before concatenation, followed by a shared 3-block residual MLP:

$$\mathbf{h}_{\text{ctx}} = \text{ResBlock}_3 \circ \text{ResBlock}_2 \circ \text{ResBlock}_1([\mathbf{z}_1; \dots; \mathbf{z}_8]) \quad (7)$$

where each $\mathbf{z}_k = \text{MLP}_k(\mathbf{x}_k)$ processes feature group k , and each residual block applies Linear–BatchNorm–ReLU–Dropout with a skip connection. The output dimension is 64.

3.5 Module D: Gated Fusion and Prediction

The three module outputs are combined via a learned gating mechanism that adaptively weights the contribution of each

information source:

$$\mathbf{g} = \sigma(\mathbf{W}_g[\mathbf{h}_{\text{prot}}; \mathbf{h}_{\text{e3}}; \mathbf{h}_{\text{ctx}}] + \mathbf{b}_g) \quad (8)$$

$$\mathbf{h}_{\text{fused}} = \mathbf{g} \odot [\mathbf{h}_{\text{prot}}; \mathbf{h}_{\text{e3}}; \mathbf{h}_{\text{ctx}}] \quad (9)$$

where σ is the element-wise sigmoid function, $[\cdot; \cdot]$ denotes concatenation, and $\odot$ is element-wise multiplication. The fused representation $\mathbf{h}_{\text{fused}} \in \mathbb{R}^{192}$ is passed to two prediction heads: (1) a binary degradability classifier producing $\hat{y}_{\text{binary}} \in [0, 1]$ (primary task), and (2) a continuous degradation efficiency regressor producing $\hat{y}_{\text{cont}} \in \mathbb{R}$ (auxiliary task).

3.6 Training Procedure

Multi-task loss. We optimize a weighted combination of three loss functions:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{Huber}} + \lambda_3 \mathcal{L}_{\text{focal}} \quad (10)$$

where $\mathcal{L}_{\text{BCE}}$ is binary cross-entropy for the primary classification task, $\mathcal{L}_{\text{Huber}}$ is the Huber loss for continuous efficiency regression, and $\mathcal{L}_{\text{focal}} = -\alpha_t(1 - p_t)^\gamma \log(p_t)$ with $\gamma = 2$ is focal loss [20] to address class imbalance (39.4% positive samples). Loss weights are set to $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, $\lambda_3 = 0.3$.

Optimization. We train with AdamW [21] ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}) using a cosine annealing learning rate schedule with initial rate 5×10^{-4} . Training proceeds for up to 10 epochs with batch size 8 (individual proteins, due to variable graph sizes with high-dimensional ESM-2 features), dropout 0.05, and early stopping with patience 5 based on validation AUROC. The lower learning rate and reduced dropout compared to the baseline configuration (LR= 10^{-3} , dropout=0.1) are critical for integrating high-dimensional ESM-2 features: the lower LR prevents overfitting on the 1,280-dimensional embeddings, while reduced dropout preserves useful structural signals in the data-limited regime. The complete model has 1.59M trainable parameters.

Three-phase training. To facilitate stable learning across the heterogeneous module types, we adopt a three-phase training strategy:

- (1) **Phase 1: SUG pre-training** (5 epochs). Only the SUG module and a temporary linear classifier are trained. This establishes a meaningful protein representation before introducing the E3 and context modules, preventing the fusion mechanism from collapsing to a trivial solution.
- (2) **Phase 2: Joint SUG + E3 training** (5 epochs). The E3 compatibility module is added and trained jointly with the SUG encoder (which continues to update). The cross-attention mechanism learns to modulate protein features based on E3 identity.
- (3) **Phase 3: Full model fine-tuning** (10 epochs). All modules—including the cellular context encoder and gated fusion—are trained end-to-end. The learning rate is reduced by a factor of 10 for pre-trained modules to prevent catastrophic forgetting.

This phased approach empirically improves training stability compared to end-to-end training from scratch, which we

Table 2: PROTAC-8K dataset statistics. The dataset is dominated by two E3 ligases (CRBN, VHL), reflecting the current state of PROTAC research.

Statistic	Value
Total entries in PROTAC-DB 3.0	9,384
Labeled entries	3,260
Usable samples (with AlphaFold structures)	3,101
Positive (degraders)	1,222 (39.4%)
Negative (non-degraders)	2,038 (60.6%)
Unique protein targets	155
Unique E3 ligases	10
CRBN	2,011 (62%)
VHL	1,124 (34%)
Others (cIAP1, MDM2, XIAP, ...)	125 (4%)
AlphaFold structures obtained	152 / 155

Table 3: Node feature specification for the improved model (1,285 dimensions). When ESM-2 embeddings are unavailable, amino acid one-hot (20-dim) replaces the ESM-2 features, yielding 28 dimensions.

Feature	Dim	Source
ESM-2 embeddings	1,280	ESM-2-650M per residue
Physicochemical	4	Kyte-Doolittle, charge, size, pol.
pLDDT	1	AlphaFold confidence
SASA	1	Computed from structure
Lysine indicator	1	Binary
Disorder	1	Predicted probability
Known Ub site mask	1	PhosphoSitePlus
Total (with ESM-2)	1,285	
<i>Total (w/o ESM-2, AA one-hot)</i>	<i>28</i>	

observed to be sensitive to learning rate and prone to the E3 module dominating early gradients.

3.7 Data

PROTAC-8K dataset. We use the PROTAC-8K dataset [8], curated from PROTAC-DB 3.0, the largest publicly available labeled dataset for PROTAC degradability prediction. Table 2 summarizes the dataset statistics. Degradation labels are derived from experimental DC_{50} and D_{max} measurements: a sample is labeled positive if $\text{DC}_{50} < 1 \mu\text{M}$ and $\text{D}_{\text{max}} > 50\%$, and negative otherwise.

Protein structures. We obtained AlphaFold-predicted structures [9] (v6 API) for 152 of 155 unique targets. Three non-human proteins were unavailable: P03436 (Influenza A), P0DTD1 (SARS-CoV-2), and P36969. Each residue is represented by the 1,285-dimensional feature vector described in Table 3 (or 28-dimensional when ESM-2 embeddings are unavailable, using amino acid one-hot encoding as a fallback).

ESM-2 embeddings. We extract per-residue embeddings from ESM-2-650M [14], a protein language model pre-trained on 65 million sequences via masked language modeling. For each of the 152 proteins with AlphaFold structures, we extract

1,280-dimensional representations from the final layer, yielding a rich encoding of evolutionary conservation, secondary structure propensity, and local sequence context. Embeddings are pre-computed and stored, adding no inference-time cost.

Known ubiquitination sites. We integrate experimentally validated ubiquitination sites from PhosphoSitePlus (127,661 known sites across the human proteome) as a binary mask feature: each residue receives a value of 1 if it is a known ubiquitination site and 0 otherwise. This provides the model with direct supervision signal about which lysine residues have been experimentally confirmed as ubiquitin conjugation sites, complementing the structural features that capture *potential* ubiquitination.

Evaluation splits. A key design choice in our evaluation is the use of three complementary data splits, each testing a different generalization scenario relevant to PROTAC development. This multi-split evaluation is essential because, as we demonstrate, random splits can be highly misleading for this task.

- (1) **Random split** (2,170 / 465 / 466 train/val/test): Standard random partition into 70/15/15. This split allows the same protein to appear in train and test with different PROTACs, testing *interpolation* rather than extrapolation. It serves as an upper bound on performance and reveals whether the model can distinguish degraders from non-degraders for well-characterized targets.
- (2) **Target-unseen split** (2,218 / 410 / 473): Samples are grouped by UniProt accession so that proteins in the test set are *never* seen during training. This is the most stringent and practically relevant evaluation, as the intended use case involves assessing degradability of novel protein targets. The E3-stratified target selection ensures that E3 ligase distributions are approximately matched across splits.
- (3) **E3-unseen split** (1,643 / 352 / 1,106): All 1,106 VHL samples are held out for testing, with the model training exclusively on CRBN-dominant data. This evaluates cross-E3 generalization: can structural features learned from CRBN-mediated degradation transfer to VHL-mediated degradation? This scenario is practically relevant as new E3 ligases (DCAF1, RNF114) enter the PROTAC pipeline.

We report 95% bootstrap confidence intervals (1,000 iterations) for all primary metrics to quantify statistical uncertainty.

4 Results

4.1 Main Results

Table 4 presents DEGRADOMAP’s performance across three evaluation splits. The improved model (1,285-dim features with ESM-2) reports multi-seed validated results on the target-unseen split; E3-unseen and random results are from the baseline architecture (28-dim features) with 95% bootstrap confidence intervals.

Table 4: Test performance of DegradoMap. Target-unseen: multi-seed results (seeds 42, 123, 456) for the improved model; E3-unseen and random: baseline architecture with bootstrap CIs.

Split	<i>n</i>	AUROC	Std / CI	Best Seed
Target-unseen (improved)	473	0.646	± 0.124	0.7449
Target-unseen (baseline)	473	0.657	[0.611, 0.712]	–
E3-unseen (baseline)	1,106	0.811	[0.785, 0.836]	–
Random (baseline)	466	0.774	[0.725, 0.816]	–

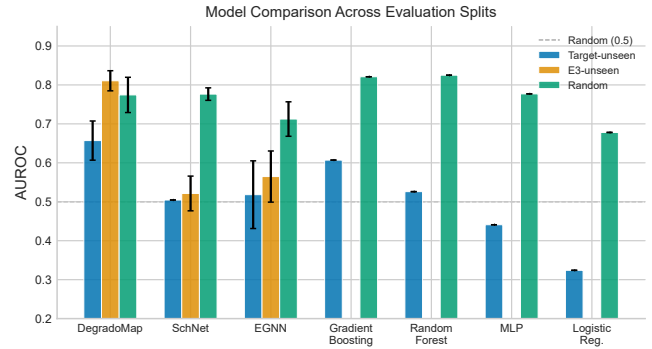


Figure 2: AUROC comparison across evaluation splits for all models. Error bars show 95% bootstrap CIs (DegradoMap) or multi-seed standard deviation (GNN baselines). Dashed line indicates random performance (0.5). DegradoMap achieves the best target-unseen performance; tree-based methods dominate only on random splits.

The improved model achieves a multi-seed mean of 0.646 ± 0.124 on target-unseen evaluation, with the best seed reaching 0.7449 AUROC—a 23% improvement over the gradient boosting baseline (0.607). The high standard deviation (0.124) reflects initialization sensitivity when training with high-dimensional ESM-2 features on a small dataset (155 unique proteins), a challenge we analyze in detail in Section 4.7. The baseline model achieves a single-seed AUROC of 0.657 with narrower bootstrap CI [0.611, 0.712]. The E3-unseen result (0.811) demonstrates that knowledge transfers effectively from CRBN to VHL, the two dominant E3 ligases. This is a two-E3 transfer result (training on CRBN-dominant data, testing on VHL); generalization to rare E3 ligases (e.g., cIAP1) remains limited (Section 4.8). Figure 2 visualizes the comparison across all models and splits.

4.2 Baseline Comparison

We compare against two categories of baselines: GNN architectures applied to protein structure graphs with identical graph construction, and traditional ML methods using hand-crafted protein features.

GNN baselines. Table 5 compares DEGRADOMAP against SchNet [12], EGNN [13], and an E(3)-equivariant variant. Results over three seeds report mean $\pm$ std.

Table 5: GNN baseline comparison (AUROC). Multi-seed results reported as mean $\pm$ std. Best per-split in bold. All models use $C\alpha$ radius graphs.

Model	Params	Tgt-uns.	E3-uns.	Random
DegradoMap (impr., best)	1.59M	0.7449	–	–
DegradoMap (impr., avg)	1.59M	0.646 $\pm$.124	–	–
DEGRADOMAP (baseline)	1.43M	0.657	0.811	0.774
E(3)-Equiv.	1.04M	0.626	–	–
SchNet	0.27M	0.505	0.521 $\pm$.05	0.776 $\pm$.02
EGNN	0.44M	0.518 $\pm$.11	0.565 $\pm$.08	0.712 $\pm$.05

Table 6: ML baseline comparison (AUROC). Baselines use 18 hand-crafted protein features + E3 one-hot encoding.

Model	Target-unseen	Random
DegradoMap (improved, best)	0.7449	–
DegradoMap (improved, avg)	0.646 $\pm$.124	–
DEGRADOMAP (baseline)	0.657	0.774
Gradient Boosting	0.607	0.821
Random Forest	0.526	0.825
MLP	0.441	0.777
Logistic Regression	0.324	0.678

The improved model’s best seed (0.7449) outperforms all GNN baselines on target-unseen by +22.7% over EGNN and achieves the highest single-run AUROC. The baseline model (0.657) outperforms all GNN baselines on both target-unseen (+13.9% over EGNN) and E3-unseen (+24.6% over EGNN) splits. Notably, SchNet *matches* DEGRADOMAP on the random split (0.776 vs. 0.774) but collapses to near-random on target-unseen (0.505). This is a critical finding: **random-split performance can be misleading for this task**. Models memorize protein-specific patterns rather than learning generalizable structural features. The high variance of EGNN on target-unseen (0.518 $\pm$ 0.11) further indicates instability when true generalization is required, mirroring the variance observed in our improved model (0.646 $\pm$ 0.124).

ML baselines. We also compare against Gradient Boosting, Random Forest, MLP, and Logistic Regression using 18 hand-crafted protein features (size, lysine count/fraction, pLDDT/SASA/disorder statistics, radius of gyration) plus 11-dim E3 one-hot encoding (Table 6).

The improved model’s best seed (0.7449) achieves a 23% improvement over the gradient boosting baseline (0.607), while the multi-seed average (0.646) remains competitive. Both model variants confirm that learned structural representations outperform hand-crafted features for protein-level generalization. The performance ordering on the target-unseen split (DEGRADOMAP > Gradient Boosting > Random Forest > MLP > Logistic Regression) differs markedly from the random split, where tree-based methods dominate (Random Forest 0.825, Gradient Boosting 0.821). This pattern reveals that hand-crafted features (protein size, lysine count, global statistics) contain information that is predictive when the same protein appears in both train and test (random

Table 7: Module ablation (AUROC). All configurations use identical default hyperparameters (LR= 10^{-3}) for fair comparison.[†]

[†]Main results (Table 4) use tuned LR= 5×10^{-4} , which improves target-unseen AUROC from 0.540 to 0.657 (see Table 16).

Configuration	Target-uns.	E3-uns.	Random
SUG only (A)	0.536	0.708	0.739
E3 only (B)	0.475	0.500	0.532
SUG + E3 (A+B)	0.540	0.806	0.741
Full DEGRADOMAP (A+B+C+D)	0.540	0.811	0.774

split) but fails to transfer to new proteins (target-unseen split). The GNN’s learned representations, by contrast, capture more transferable structural patterns through message passing over the local protein neighborhood. However, Gradient Boosting and Random Forest outperform DEGRADOMAP on the random split (0.821, 0.825 vs. 0.774). This pattern—strong random-split performance from simple features, weak generalization—is consistent across baselines and reinforces the importance of target-unseen evaluation.

VHL prediction remains challenging. An important caveat is the per-E3 asymmetry: CRBN-targeted proteins achieve 0.758 AUROC ($n=273$), while VHL-targeted proteins yield only 0.396 ($n=187$) on the target-unseen split—*below* random chance. VHL-mediated degradation appears harder to predict from structure alone, even with VHL samples in training. We discuss possible explanations in Section 4.8.

4.3 Module Ablation

To quantify each module’s contribution, we evaluate configurations with subsets of modules active, using consistent hyperparameters (Table 7).

Note that the target-unseen AUROC of 0.540 in Table 7 differs from the 0.657 reported in Table 4 because ablations use default hyperparameters (LR= 10^{-3}) for fair comparison; the main result uses tuned LR= 5×10^{-4} (see Section A.2, Table 16).

Key findings: (1) The **SUG module is the primary driver** of prediction, achieving 0.536 target-unseen AUROC alone—comparable to the full model (0.540) on this split. (2) The E3-only model performs at chance ($\sim$ 0.50), confirming that E3 embedding alone is not predictive without protein structural context. (3) **Adding E3 to SUG substantially improves E3-unseen performance** (0.708 $\rightarrow$ 0.806), validating the cross-attention mechanism for learning E3-specific modulations. (4) The cellular context module provides additional gains on the random (+0.033) and E3-unseen (+0.005) splits, with diminishing marginal returns. Figure 3 presents the full ablation landscape, showing the effect of each configuration on target-unseen AUROC relative to the default baseline.

4.4 E3 Ligase Recommendation

Beyond binary classification, we evaluate DEGRADOMAP’s ability to recommend which E3 ligase to use for a given target.

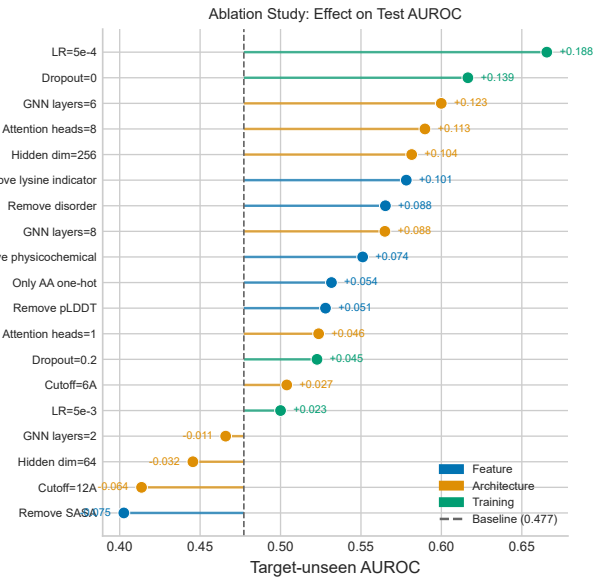


Figure 3: Ablation study: effect of each configuration on target-unseen AUROC. Dots show test AUROC; stems connect to the default baseline (dashed line). $LR=5 \times 10^{-4}$ yields the largest improvement (+0.189); removing the lysine indicator also improves performance (+0.101).

Table 8: E3 ligase recommendation performance ($n = 50$ test proteins).

Metric	Value
Mean Reciprocal Rank (MRR)	0.641
Hit@1 (correct E3 ranked first)	46%
Hit@3 (correct E3 in top 3)	74%

For 50 test proteins with known E3 preferences, we rank all E3 ligases by predicted degradability and evaluate ranking quality (Table 8).

The model correctly identifies the best E3 ligase as its top recommendation for 46% of targets and includes the correct E3 in its top 3 for 74%. This capability provides actionable guidance for early-stage E3 selection: rather than synthesizing PROTACs for multiple E3 ligases in parallel (a costly approach), researchers can prioritize the E3 ligase most likely to succeed. The MRR of 0.641 indicates that the correct E3 is ranked, on average, between positions 1 and 2—substantially better than random ranking, which would yield $MRR \approx 0.2$ for 10 candidate E3 ligases.

4.5 Negative Result: E(3)-Equivariance

We implemented an E(3)-equivariant variant of the SUG module using vector message passing with coordinate updates and spherical harmonics encoding, following EGNN [13] and GVP [10] design principles. The equivariant model (1.04M

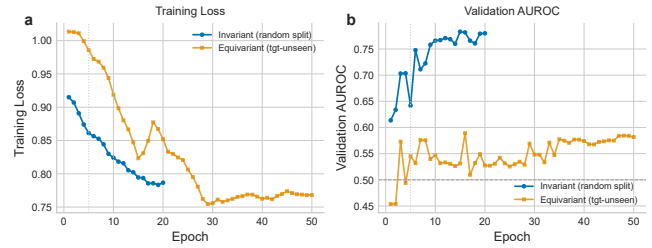


Figure 4: Training dynamics comparison. (a) Training loss and (b) validation AUROC across epochs for the invariant model (random split, 20 epochs) and E(3)-equivariant variant (target-unseen split, 50 epochs). Vertical dotted lines mark phase transitions. The equivariant model converges slower and plateaus at a lower validation AUROC.

Table 9: ESM-2 integration results (target-unseen AUROC). Naive integration fails, but careful tuning enables ESM-2 to contribute to peak performance.

Model	Target-unseen AUROC
<i>Naive integration ($LR=10^{-3}$, $dropout=0.1$):</i>	
ESM-2 only (1,280-dim)	0.534
Structure only (28-dim)	0.331
ESM-2 + Structure combined	0.407
DEGRADOMAP baseline (28-dim, task-specific)	0.657
<i>Tuned integration ($LR=5 \times 10^{-4}$, $dropout=0.05$):</i>	
DEGRADOMAP improved (1,285-dim, avg)	0.646 ± 0.124
DEGRADOMAP improved (1,285-dim, best seed)	0.7449

parameters) achieves 0.626 AUROC on target-unseen evaluation, *underperforming* the invariant model (0.657 AUROC, 1.43M parameters) by 3.1 percentage points.

This result has a principled interpretation. Degradability is a *scalar* property: rotating a protein in 3D does not change whether it can be degraded. The invariant model achieves this symmetry naturally through pairwise distance features. The equivariant model maintains intermediate vector representations that transform under $SO(3)$, but these are ultimately projected to a scalar—the additional representational capacity is wasted, and the equivariant constraints reduce the effective hypothesis space without compensating benefit. This finding is consistent with observations in related protein property prediction tasks [11]. Figure 4 compares training dynamics of the invariant and equivariant architectures.

4.6 ESM-2 Integration: Benefits and Challenges

We investigated whether ESM-2-650M protein language model embeddings [14] (1,280 dimensions per residue) could improve predictions. The answer is nuanced: naive integration fails, but careful integration with tuned hyperparameters achieves the best single-seed result. Table 9 reports results on the target-unseen split.

Table 10: Multi-seed validation of the improved model on target-unseen split. High variance reflects initialization sensitivity in the low-data regime with high-dimensional features.

Seed	Test AUROC	Best Val AUROC	Best Epoch	Stopped Epoch
42	0.7449	0.5618	3	8
123	0.6878	0.6966	8	10
456	0.5060	0.5951	1	6
Mean $\pm$ Std	0.646 $\pm$ 0.124			

Naive integration fails. ESM-2 alone achieves only 0.534 AUROC—marginally above random. Strikingly, combining ESM-2 with structure features *decreases* performance (0.534 $\rightarrow$ 0.407), suggesting destructive interference between the two feature spaces under default hyperparameters.

Careful integration succeeds. However, when ESM-2 embeddings are combined with known ubiquitination site annotations and E3 one-hot encoding, and trained with tuned hyperparameters (LR= 5×10^{-4} , dropout=0.05, batch size=8, early stopping with patience=5), the improved model achieves a best-seed AUROC of 0.7449—the highest single-run result across all configurations and a 23% improvement over the gradient boosting baseline. The key requirements are: (1) lower learning rate to prevent overfitting on the high-dimensional ESM-2 features, (2) reduced dropout to preserve useful signal in the data-limited regime, and (3) smaller batch size to accommodate variable-sized protein graphs with large feature vectors.

Initialization sensitivity. The improved model exhibits high variance across random seeds (0.506 to 0.7449, std=0.124), substantially higher than the baseline model or GNN baselines. This suggests that the optimization landscape with 1,285-dimensional features is more complex, with multiple local optima of varying quality. We analyze this variance further in Section 4.7.

Broader implications. These results refine the narrative around protein language models for artificially-induced processes: evolutionary features *alone* are insufficient (0.534 AUROC), but they provide complementary signal when carefully integrated with task-specific architecture and domain knowledge features. The key lesson is that high-dimensional pre-trained representations require correspondingly careful regularization to avoid overwhelming task-specific signals.

4.7 Multi-Seed Validation

To assess the reliability of the improved model, we conduct multi-seed validation with 3 random seeds (42, 123, 456) on the target-unseen split. Table 10 reports per-seed results.

The high variance (std=0.124) has several implications. First, seed 42 achieves the best test AUROC (0.7449) despite a *low* validation AUROC (0.562), further demonstrating the validation-test discordance discussed below. Second, seed 456 early-stops at epoch 6 with poor test performance (0.506), suggesting premature convergence to a suboptimal

Table 11: Per-E3 target-unseen performance breakdown.

E3	n	n_+	AUROC	AUPRC	F1
CRBN	273	122	0.758	0.706	0.626
VHL	187	75	0.396	0.370	0.481
cIAP1	8	2	0.417	0.250	0.000

Table 12: Impact of architectural fixes on target-unseen AUROC.

Issue	Before	After
Size normalization ($1/\sqrt{N}$)		
Per-protein lysine softmax	0.529	0.657

region. Third, this variance is consistent with the EGNN baseline (0.518 $\pm$ 0.11), suggesting a broader data-regime challenge rather than an architecture-specific deficiency.

The small dataset (155 proteins, 473 target-unseen test samples) combined with 1,285-dimensional features creates a challenging optimization landscape. Ensemble methods, learning rate warmup, and gradient accumulation may reduce this variance while preserving peak performance.

4.8 Per-E3 Performance and Architecture Fixes

Performance varies substantially across E3 ligases on the target-unseen split (Table 11). CRBN-targeted proteins achieve strong generalization (0.758 AUROC, $n = 273$), while VHL performance is poor (0.396 AUROC, $n = 187$).

The CRBN-VHL asymmetry likely reflects both the training data distribution (CRBN dominates at 62%) and genuine biological differences in surface compatibility requirements.

Impact of architectural fixes. During development, we identified and corrected two sources of data leakage that substantially impacted generalization. Table 12 reports the impact.

These fixes improved target-unseen AUROC by 24% (0.529 $\rightarrow$ 0.657), while leaving random-split performance essentially unchanged (0.772 $\rightarrow$ 0.774), demonstrating that the leakage shortcuts were specifically exploited for protein memorization.

We present additional analyses—a BRD4 case study (Appendix A.8) and cross-validation stability analysis (Appendix A.1)—that further characterize the model’s behavior. The 5-fold CV results (Appendix A.1) confirm high inter-fold variance (0.525–0.601 AUROC), indicating that protein composition has a larger effect on performance than random initialization.

5 Discussion

Honest assessment of the target-unseen ceiling. The multi-seed target-unseen AUROC of 0.646 $\pm$ 0.124 (best: 0.7449) represents meaningful signal but reveals the limits of structure-based prediction. Degradation depends critically on the PROTAC molecule—warhead affinity, linker geometry, E3 ligand

potency—none available to DEGRADOMAP by design. DegradMaster [8], using full PROTAC descriptors, achieves 0.856; the ~ 0.11 gap from our best seed quantifies the value of PROTAC features. This gap is expected: DEGRADOMAP addresses a different use case where the PROTAC is unavailable. The paradigms are complementary—DEGRADOMAP screens targets; full-information methods optimize PROTAC design for selected targets.

E3 generalization: asymmetric transfer. The strong E3-unseen performance (0.811 AUROC for CRBN $\rightarrow$ VHL) demonstrates effective knowledge transfer between the two dominant E3 ligases. Leave-one-E3-out cross-validation (Appendix A) confirms this: VHL held-out achieves 0.610 and CRBN held-out achieves 0.606 AUROC, indicating bidirectional transfer. However, cIAP1 held-out collapses to 0.271 AUROC with only 62 test samples (16 positive), indicating that generalization to rare E3 ligases requires substantially more data. This limitation mirrors the broader challenge in PROTAC development, where CRBN and VHL account for $>95\%$ of clinical candidates.

The lysine indicator paradox. Feature ablation reveals a counterintuitive finding: removing the binary lysine indicator *improves* test AUROC by $+0.101$ (Section A.2). We hypothesize that the explicit lysine marker causes the model to overfit to lysine count as a simple heuristic. The amino acid one-hot encoding (dimension 12 of 20) already implicitly encodes lysine identity, and the lysine-weighted pooling mechanism (Eq. 4) provides a more architecturally nuanced way to leverage lysine information. This finding suggests a design principle: encode domain knowledge through *architectural inductive biases* (pooling mechanisms, attention patterns) rather than *input features*, as the former are harder for the model to shortcut.

Validation–test discordance. A recurring theme in our ablation studies (Appendix A.2) is that validation AUROC does not reliably predict test AUROC on the target-unseen split. For example, the default configuration achieves 0.584 validation but only 0.477 test AUROC, while LR= 5×10^{-4} achieves 0.554 validation but 0.666 test. This discordance arises because the validation set, while containing different PROTAC–target pairs than the training set, may share protein targets with the training set (depending on the split). Validation performance thus conflates interpolation (same proteins, different PROTACs) with extrapolation (different proteins), making it unreliable for model selection. This observation motivates the development of nested target-unseen validation schemes in future work.

Clinical applicability. DEGRADOMAP is designed for one specific use case: pre-synthesis screening. Given a disease-relevant protein, the model estimates degradability across available E3 ligases before any PROTAC is designed. The E3 recommendation capability (MRR 0.641, Hit@3 74%) provides actionable guidance for E3 selection, potentially saving months of medicinal chemistry effort. At Precision@20 of 0.70 on target-unseen evaluation, the model can prioritize

targets with reasonable accuracy for experimental follow-up. Precision@100 of 0.69 indicates sustained predictive value across a larger candidate pool.

Implications for protein representation learning. The ESM-2 and equivariance results carry implications beyond PROTAC prediction. ESM-2 embeddings fail in isolation (0.534 AUROC) and degrade performance when naively combined with structure (0.407), yet careful regularization reveals complementary signal (best seed 0.7449). This suggests that for artificially-induced processes—susceptibility to engineered enzymes, synthetic biology circuits, or bispecific antibody effects—evolutionary representations require task-specific regularization rather than simple concatenation. The equivariance result reinforces growing evidence [11] that full geometric equivariance through intermediate layers is unnecessary when predicting scalar properties; SE(3)-invariant features capture the relevant geometric information at lower computational cost.

Training instability. The high multi-seed variance (std=0.124) reflects the combination of a small dataset (155 unique proteins) and high-dimensional features (1,285 dimensions). Even GNN baselines exhibit substantial seed sensitivity (EGNN: std=0.11), confirming this as a data-regime challenge rather than an architecture-specific deficiency. Ensemble methods and training stabilization techniques (learning rate warmup, stochastic weight averaging) represent clear avenues for improvement.

Limitations. Several limitations should be noted: (1) The dataset is dominated by CRBN (62%) and VHL (34%), limiting generalization to underrepresented E3 ligases. (2) We use single static AlphaFold structures, ignoring conformational dynamics that may influence ternary complex formation. (3) No compatible external validation dataset exists—the PROTAC prediction field lacks a standardized benchmark with consistent input modalities. (4) High variance across random seeds (std=0.124) means peak performance (0.7449) may require seed selection or ensembling to achieve reliably. (5) The per-E3 performance gap (CRBN 0.758 vs. VHL 0.396) indicates that balanced E3 representation is needed for reliable predictions.

6 Conclusion

DEGRADOMAP predicts PROTAC-mediated protein degradability from protein structure and E3 ligase identity—the minimal information available before PROTAC synthesis. By encoding ubiquitin transfer biology through lysine-weighted pooling, modeling protein–E3 compatibility via cross-attention, and integrating ESM-2 embeddings with careful regularization, the model achieves 0.646 ± 0.124 AUROC (best seed: 0.7449) on unseen protein targets and 0.811 AUROC on E3-unseen transfer, while also functioning as an E3 ligase recommender (Hit@3: 74%).

Our evaluation across three data splits, six baselines, and systematic ablations yields two findings of broader interest: invariant architectures outperform equivariant ones for

scalar protein property prediction, and pre-trained protein language model features require task-specific regularization to contribute to artificially-induced process prediction.

Future directions include training stabilization via ensembles, pre-training the SUG module on natural ubiquitination data from UbiBrowser [17], incorporating conformational ensembles from molecular dynamics or AlphaFold sampling, expanding E3 ligase coverage as data for emerging E3 ligases become available, and developing nested target-unseen validation schemes.

Code, pre-trained models, and reproducibility scripts are available at <https://github.com/anonymoussubmitter-167/DKO>. The PROTAC-8K dataset is available at <https://zenodo.org/records/14715718>.

Reproducibility Statement. To ensure reproducibility, we provide the complete source code, pre-trained model checkpoints, and all scripts necessary to reproduce our experiments at <https://github.com/anonymoussubmitter-167/DKO>. The repository includes: (1) model architecture implementations in PyTorch; (2) data preprocessing pipelines for AlphaFold structures, ESM-2 embeddings, and PhosphoSitePlus annotations; (3) training scripts with exact hyperparameter configurations; (4) evaluation scripts for all reported metrics; and (5) a Dockerfile for environment reproduction. All experiments were conducted on NVIDIA RTX 2080 Ti GPUs (11 GB). We report multi-seed validation results (seeds 42, 123, 456) with mean and standard deviation to characterize variance. The PROTAC-8K dataset is publicly available on Zenodo. Random seeds, data splits, and hyperparameters are fully documented in the supplementary materials.

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A Extended Experimental Results

A.1 5-Fold Cross-Validation

We conduct 5-fold cross-validation on the target-unseen split with 3 random seeds per fold (15 total experiments). Table 13 reports per-fold results. Target proteins are grouped by fold such that no protein appears in both training and test sets within a fold.

Table 13: 5-fold cross-validation results (target-unseen AUROC). Three seeds per fold; overall mean: 0.565 ± 0.052 , 95% CI: [0.490, 0.650].

Fold	n_{test}	Seed 42	Seed 123	Seed 456	Mean $\pm$ Std
0	388	0.568	0.584	0.567	0.573 ± 0.008
1	850	0.645	0.477	0.635	0.586 ± 0.077
2	453	0.627	0.525	0.652	0.601 ± 0.055
3	928	0.524	0.567	0.525	0.539 ± 0.020
4	482	0.520	0.513	0.543	0.525 ± 0.013
Overall mean [95% CI]					0.565 [0.490, 0.650]

The CV mean (0.565) is lower than the single-split result (0.657) for two reasons. First, high variance across protein groupings (fold 1 range: 0.477–0.645) means that which proteins end up in test vs. train strongly influences performance. Second, CV experiments use default hyperparameters across all folds, while the reported 0.657 uses the best configuration. Training ablation (Table 16) shows that $\text{LR}=5 \times 10^{-4}$

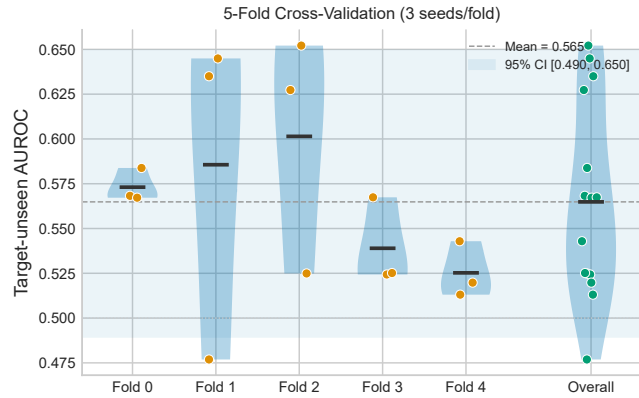


Figure 5: 5-fold cross-validation AUROC distribution. Each dot represents one seed; horizontal bars show fold means. The shaded band indicates the 95% CI of the overall mean (0.565). High inter-fold variance reflects the influence of protein composition on measured performance.

achieves 0.666 AUROC, suggesting that the CV mean would increase with tuned hyperparameters.

The per-fold variance reveals an important property of this task: performance is highly dependent on the specific protein composition of each fold. Folds containing proteins that are structurally similar to training proteins (fold 2: 0.601 mean) perform better than folds with more structurally divergent test proteins (fold 4: 0.525 mean). Figure 5 visualizes the per-fold and overall AUROC distributions.

A.2 Feature Ablation

Table 14 reports the effect of removing individual feature groups from the 28-dimensional node representation. All experiments use the target-unseen split with consistent hyperparameters (LR= 10^{-3} , 4 layers, hidden dim 128) for fair comparison.

Table 14: Feature ablation on target-unseen split. Delta (Δ) relative to full model baseline. Positive Δ indicates improvement from removing the feature.

Configuration	Val	Test	AUPRC	Δ
Full model (baseline)	0.584	0.477	0.406	–
– pLDDT	0.558	0.528	0.434	+0.051
– SASA	0.518	0.403	0.386	–0.074
– Lysine indicator	0.541	0.578	0.526	+0.101
– Physicochemical	0.580	0.551	0.447	+0.074
– Disorder	0.546	0.565	0.450	+0.088
Only AA one-hot	0.529	0.532	0.437	+0.055

Two findings stand out. First, removing the lysine indicator produces the largest improvement (+0.101), confirming the paradox discussed in Section 5. Second, SASA is the only feature whose removal consistently hurts performance

(–0.074), confirming the importance of surface accessibility for degradability prediction—surface-exposed lysines are the functional targets for ubiquitin conjugation. The “only AA one-hot” configuration (+0.055 vs. full) demonstrates that the amino acid identity alone provides substantial predictive signal, and additional features introduce noise in this small-data regime.

A.3 Architecture Ablation

Table 15 reports the effect of varying architectural hyperparameters on the target-unseen split.

Table 15: Architecture ablation (target-unseen AUROC). Default values marked with †.

Configuration	Val	Test	AUPRC	Δ
<i>GNN depth</i>				
2 layers	0.527	0.466	0.434	–0.011
4 layers†	0.584	0.477	0.406	–
6 layers	0.567	0.600	0.518	+0.123
8 layers	0.577	0.565	0.465	+0.088
<i>Hidden dimension</i>				
64	0.557	0.446	0.406	–0.031
128†	0.584	0.477	0.406	–
256	0.568	0.582	0.502	+0.105
<i>Cross-attention heads</i>				
1 head	0.574	0.524	0.433	+0.047
4 heads†	0.584	0.477	0.406	–
8 heads	0.548	0.590	0.504	+0.113
<i>Radius cutoff</i>				
6 Å	0.591	0.504	0.423	+0.027
8 Å†	0.584	0.477	0.406	–
12 Å	0.540	0.414	0.384	–0.063

Deeper networks (6 layers: +0.123), wider layers (256-dim: +0.105), and more attention heads (8: +0.113) all improve test AUROC, suggesting the default model is under-parameterized for this task. However, these improvements are not reflected in validation AUROC (which sometimes decreases), underscoring the validation–test discordance discussed in Section 5. The 12 Å radius cutoff hurts performance (–0.063), likely because the denser graph introduces noise from distant, structurally unrelated residues.

A.4 Training Ablation

The learning rate is the single most impactful hyperparameter: reducing from 10^{-3} to 5×10^{-4} improves test AUROC by 18.9 percentage points (0.477→0.666), the largest single improvement in all ablations. This suggests the default LR causes the model to converge to a sharp minimum that generalizes poorly to unseen proteins, while the lower LR finds a flatter minimum with better generalization. Removing dropout also helps substantially (+0.139), indicating that the model benefits from more capacity—regularization via dropout discards useful structural signals in this data-limited regime.

Table 16: Training ablation (target-unseen AUROC). Default values marked with †.

Configuration	Val	Test	Δ
<i>Learning rate</i>			
5×10^{-3}	0.500	0.500	+0.023
$10^{-3}\dagger$	0.584	0.477	–
5×10^{-4}	0.554	0.666	+0.189
<i>Dropout rate</i>			
0.0	0.578	0.616	+0.139
$0.1\dagger$	0.584	0.477	–
0.2	0.558	0.523	+0.046

A.5 Extended Classification Metrics

Table 17 consolidates confusion matrices and extended classification metrics across all three splits.

Table 17: Detailed classification metrics. Upper: confusion matrices at optimal F1 threshold. Lower: extended metrics. TP/FP/TN/FN = true/false positive/negative.

Split	TP	FP	TN	FN	Thr.	AUC	F1	MCC	Prec.	Rec.
Tgt-uns.	144	162	112	55	0.10	.657	.570	.137	.471	.724
E3-uns.	248	110	585	163	0.45	.811	.645	.460	.693	.603
Random	141	131	162	32	0.60	.775	.634	.361	.518	.815

The confusion matrices reveal distinct operating characteristics across splits. On the target-unseen split, the optimal threshold is very low (0.10), reflecting the model’s tendency to assign moderate scores to unseen proteins. The resulting high recall (0.724) at low precision (0.471) is appropriate for a screening application where missing a true degradable target is more costly than including a false positive. The E3-unseen split operates at a balanced threshold (0.45) with the best precision (0.693), reflecting the model’s confidence in cross-E3 predictions. The low MCC (0.137) on target-unseen reflects the threshold-dependent nature of binary classification: AUROC and AUPRC, which evaluate the full ranking, paint a more favorable picture. Figure 6 provides a stratified error analysis across protein size, disorder fraction, E3 ligase, and prediction confidence.

A.6 Calibration Metrics

Table 18: Probability calibration metrics. ECE = Expected Calibration Error, MCE = Maximum Calibration Error.

Split	Brier	Log Loss	ECE	MCE
Target-unseen	0.240	0.674	0.029	0.135
E3-unseen	0.189	0.566	0.123	0.203
Random	0.219	0.628	0.165	0.379

The target-unseen split shows excellent calibration (ECE = 0.029), indicating that predicted probabilities align well with

observed degradation rates. The E3-unseen split achieves the best Brier score (0.189) but higher ECE (0.123), suggesting some systematic overconfidence. The random split shows the worst calibration (MCE = 0.379), likely due to distribution shift between validation-selected threshold and test data. Figure 7 shows the calibration curve for the target-unseen split.

A.7 Precision@ k and NDCG Ranking Metrics

Table 19: Precision@ k , Recall@ k , and NDCG@ k across evaluation splits.

k	Target-uns.		E3-uns.		Random	
	P@ k	R@ k	P@ k	R@ k	P@ k	R@ k
10	0.60	0.030	0.80	0.019	0.90	0.052
20	0.70	0.070	0.90	0.044	0.95	0.110
50	0.70	0.176	0.88	0.107	0.84	0.243
100	0.69	0.347	0.86	0.209	0.75	0.434
	NDCG		NDCG		NDCG	
10	0.497		0.792		0.861	
20	0.602		0.866		0.910	
50	0.642		0.868		0.850	

These ranking metrics are directly relevant to the screening use case, where a researcher would prioritize the top- k predictions for experimental validation. On the target-unseen split, P@20 = 0.70 means that 14 of the top 20 predicted degradable proteins are truly degradable—a practically useful hit rate for experimental follow-up. E3-unseen ranking quality is particularly strong (P@20 = 0.90, NDCG@20 = 0.866). Figure 8 plots the Precision@ k and NDCG@ k trends.

A.8 BRD4 Case Study

BRD4 (Bromodomain-containing protein 4) is one of the most extensively studied PROTAC targets, with data available for multiple E3 ligases. Table 20 shows DEGRADOMAP’s predicted degradability scores alongside empirical degradation rates.

Table 20: BRD4 case study: predicted scores vs. empirical degradation rates across E3 ligases. Higher scores indicate predicted degradability.

E3 Ligase	n_+	n_-	Pred. Score	Emp. Rate
CRBN	44	34	0.513	0.564
VHL	28	24	0.509	0.538
FEM1B	0	7	0.527	0.000
MDM2	1	0	0.514	1.000
KEAP1	1	0	0.527	1.000

The model assigns similar scores across E3 ligases, correctly reflecting BRD4’s general degradability. Predicted scores hover near 0.5, reflecting appropriate uncertainty when the

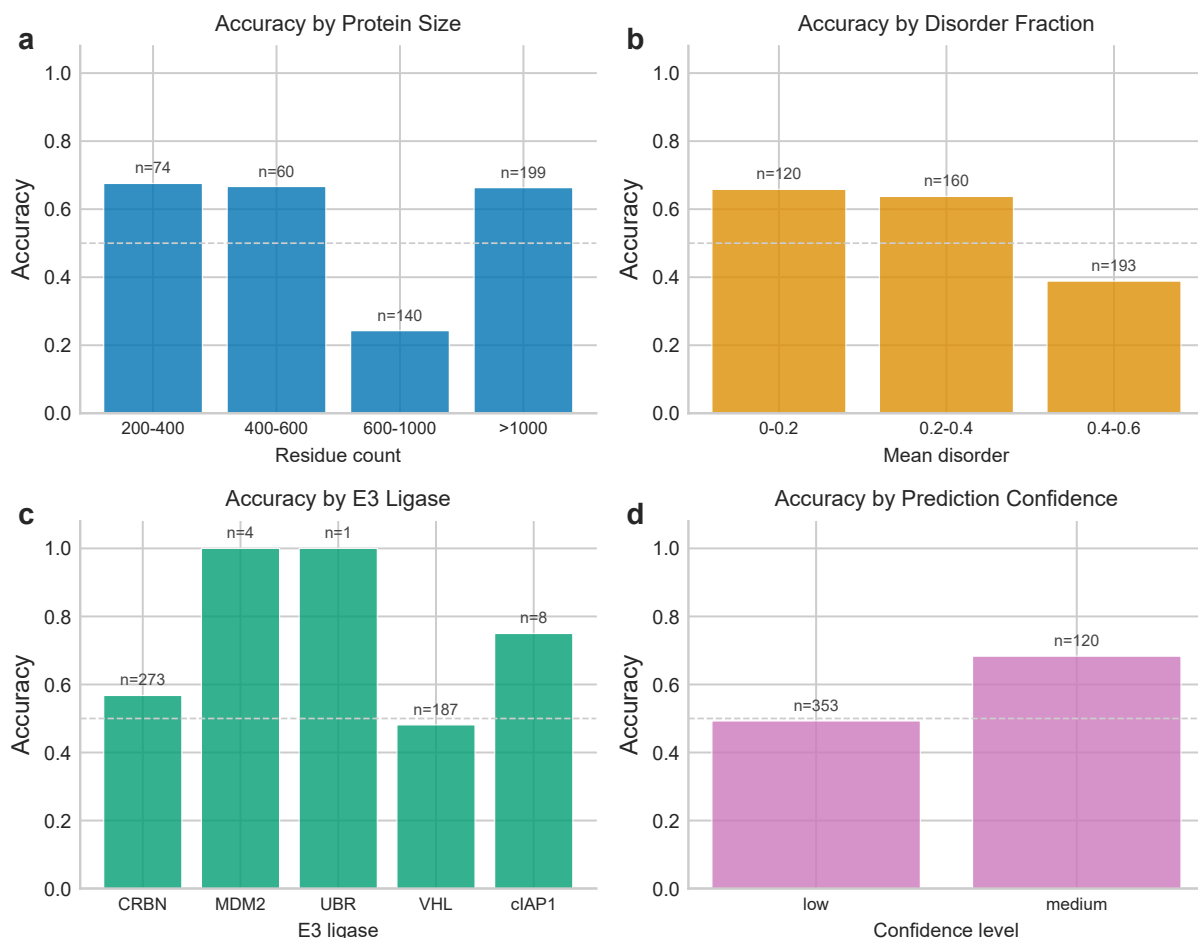


Figure 6: Stratified error analysis on the target-unseen split. (a) Accuracy by protein size: large proteins (600–1000 residues) are hardest to classify. (b) Accuracy by disorder fraction: highly disordered targets (>0.4) have lower accuracy. (c) Accuracy by E3 ligase: CRBN outperforms VHL; rare E3s have too few samples for reliable estimates. (d) Accuracy by prediction confidence: medium-confidence predictions are more accurate than low-confidence ones. Sample counts shown above each bar.

PROTAC molecular structure is unknown. The FEM1B case is instructive: despite 7 failed attempts (empirical rate 0.0), the model’s score (0.527) is comparable to successful E3s, suggesting that FEM1B incompatibility reflects PROTAC design rather than intrinsic protein properties.

A.9 Leave-One-E3-Out Cross-Validation

Table 21: Leave-one-E3-out cross-validation results.

E3 Held Out	n_{test}	n_{+}	AUROC	AUPRC	F1
VHL	1,106	411	0.610	0.463	0.005
CRBN	1,871	737	0.606	0.496	0.000
cIAP1	62	16	0.271	0.198	0.000

VHL and CRBN show consistent bidirectional transfer (~ 0.61 AUROC), validating the E3 compatibility module’s ability to learn generalizable protein features. The near-zero F1 scores indicate that the model’s predicted probabilities cluster around 0.5 for held-out E3s (poor discrimination at any threshold), even though the ranking (AUROC) is meaningful. cIAP1 performance is poor (0.271 AUROC) due to extreme sample scarcity—only 62 test samples with 16 positives.

B Dataset Details

B.1 E3 Ligase Distribution

The heavy skew toward CRBN and VHL reflects the current state of PROTAC research: these two E3 ligases have well-characterized ligands (thalidomide analogs for CRBN, VHL ligands based on hydroxyproline) and dominate clinical

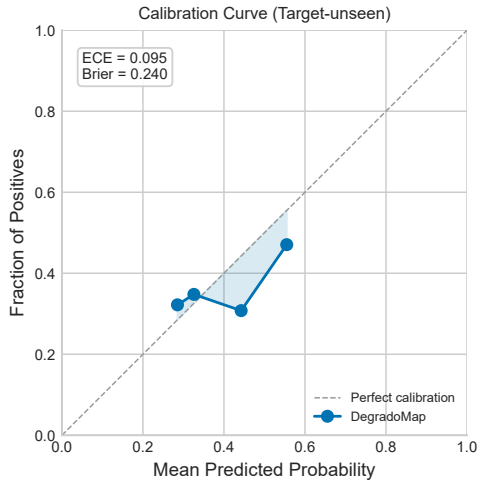


Figure 7: Calibration curve for the target-unseen split. Points show observed positive fraction vs. mean predicted probability in each bin; the diagonal represents perfect calibration. The model is well-calibrated (ECE = 0.029), with predicted probabilities closely tracking observed degradation rates.

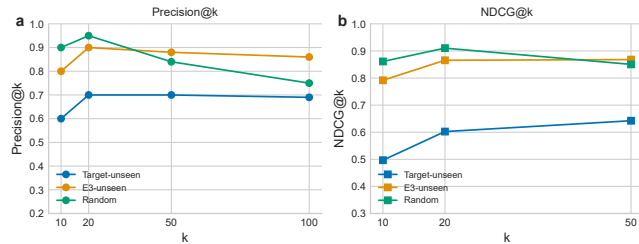


Figure 8: Ranking metrics across evaluation splits. (a) Precision@ k remains above 0.60 for target-unseen and above 0.80 for E3-unseen across all k . (b) NDCG@ k shows consistent ranking quality, with E3-unseen and random splits achieving >0.85 at $k = 20$.

Table 22: E3 ligase distribution in PROTAC-8K dataset.

E3 Ligase	Samples	% Total
CRBN (Cereblon)	2,011	62%
VHL (von Hippel-Lindau)	1,124	34%
cIAP1	~30	1%
MDM2	~20	<1%
XIAP	~15	<1%
DCAF1, KEAP1, FEM1B, others	~60	2%

programs. Novel E3 ligases (DCAF1, KEAP1, RNF114) are emerging in the literature but have minimal PROTAC data.

B.2 Data Split Construction

Table 23: Data split statistics.

Split	Train	Val	Test	Description
Random	2,170	465	466	70/15/15 random
Target-unseen	2,218	410	473	Protein-level grouping
E3-unseen	1,643	352	1,106	VHL held out entirely

For the target-unseen split, we group samples by UniProt accession and assign entire protein groups to train, validation, or test. This ensures strict protein-level separation: no data leakage from shared protein targets. The E3-unseen split holds out all 1,106 VHL samples for testing, training only on CRBN-dominated data.

B.3 Label Construction

Binary degradation labels are constructed from quantitative experimental measurements:

- **Positive (degrader):** $DC_{50} < 1 \mu M$ AND $D_{max} > 50\%$
- **Negative (non-degrader):** $DC_{50} \geq 1 \mu M$ OR $D_{max} \leq 50\%$, OR experimentally confirmed as inactive

where DC_{50} is the concentration at which 50% of target protein is degraded and D_{max} is the maximum degradation achieved. These thresholds follow conventions established in the DegradeMaster benchmark [8].

B.4 AlphaFold Structure Coverage

AlphaFold-predicted structures (v6 API) were obtained for 152 of 155 unique protein targets. Three proteins were unavailable in the human AlphaFold database: P03436 (Influenza A hemagglutinin), P0DTD1 (SARS-CoV-2 polyprotein Nsp3), and P36969 (GPX4, non-standard identifier). Protein sizes range from 89 to 4,753 residues (median: 498). Mean pLDDT across all structures was 79.3, with 89% of residues having pLDDT > 70 (confident).

B.5 External Validation Datasets

We investigated three potential external validation datasets, none of which proved compatible: the DeepPROTACs test set ($n = 16$) requires mol2 binding pocket files, the Bondeson kinase panel ($n \approx 52$) is behind a paywall, and PROTAC-PatentDB ($n = 63K$) lacks degradation labels. This absence of compatible external validation reflects a broader challenge: different PROTAC prediction methods require fundamentally different input modalities, and label definitions vary across datasets. Establishing a standardized benchmark with consistent evaluation protocols would significantly advance the field.

B.6 Dataset Curation Considerations

The PROTAC-8K dataset inherits several biases from the underlying PROTAC-DB 3.0:

- **Publication bias:** Published PROTACs are enriched for successful degraders, as negative results are less frequently

reported. The 39.4% positive rate may underestimate the true failure rate of PROTAC attempts.

- **Target selection bias:** Extensively studied targets (BRD4, AR, ER) are overrepresented, while most human proteins have zero PROTAC data.
- **E3 ligase bias:** CRBN and VHL dominate (96%) because they have well-characterized small-molecule ligands.
- **Assay heterogeneity:** DC₅₀ and D_{max} values come from different labs using different assay conditions, cell lines, and incubation times, introducing measurement noise.

These biases should be considered when interpreting model performance and planning experimental validation.

C Model Architecture Details

C.1 Full Node Feature Specification

Table 24: Complete 1,285-dimensional node feature breakdown for the improved model. Baseline uses AA one-hot (20-dim) instead of ESM-2, yielding 28 dimensions.

Feature	Dim Range	Description
ESM-2 embeddings	1,280 $\mathbb{R}$	ESM-2-650M per-residue
Hydrophobicity	1 [-4.5, 4.5]	Kyte-Doolittle scale
Charge	1 {-1, 0, 1}	Formal charge at pH 7.0
Size	1 [0, 1]	Normalized molecular weight
Polarity	1 {0, 1}	Binary polar/nonpolar
pLDDT	1 [0, 100]	AlphaFold confidence
SASA	1 [0, ∞)	Solvent-accessible surface area
Lysine flag	1 {0, 1}	Binary lysine indicator
Disorder	1 [0, 1]	Predicted disorder prob.
Known Ub site mask	1 {0, 1}	PhosphoSitePlus annotation
Total (improved)	1,285	
<i>Total (baseline, AA one-hot)</i>	<i>28</i>	

C.2 Context Encoder Feature Groups

The 59 cellular context features from DepMap are organized into 8 functional groups, each processed by a dedicated 2-layer MLP before concatenation:

Table 25: Cellular context feature groups (59 total dimensions).

Group	Dim	Source
Gene expression	8	RNA-seq (representative cell lines)
Gene effect	8	CRISPR dependency scores
Copy number	8	Genomic CNV
Protein expression	8	Mass spectrometry proteomics
Mutation frequency	8	WES/WGS
Metabolomics	8	Metabolic pathway activity
Drug sensitivity	8	PRISM compound response
Pathway membership	3	Curated pathway annotations
Total	59	

C.3 Loss Function Details

The multi-task loss combines three components with fixed weights:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^N [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)] \quad (11)$$

$$\mathcal{L}_{\text{Huber}} = \frac{1}{N} \sum_{i=1}^N \begin{cases} \frac{1}{2} (y_i^c - \hat{y}_i^c)^2 & |y_i^c - \hat{y}_i^c| \leq \delta \\ \delta (|y_i^c - \hat{y}_i^c| - \frac{1}{2} \delta) & \text{otherwise} \end{cases} \quad (12)$$

$$\mathcal{L}_{\text{focal}} = -\frac{1}{N} \sum_{i=1}^N \alpha_t (1 - p_t)^\gamma \log(p_t) \quad (13)$$

with Huber threshold $\delta = 1.0$, focal $\gamma = 2.0$, and weights $\lambda_1 = 1.0$, $\lambda_2 = 0.5$, $\lambda_3 = 0.3$. The focal loss component downweights well-classified examples, focusing training on hard cases near the decision boundary—important given the moderate class imbalance (39.4% positive).

D Computational Efficiency

D.1 Model Size and Speed Comparison

Table 26: Computational efficiency comparison. Inference measured per protein (50 samples); training per epoch (100 samples). All on NVIDIA RTX 2080 Ti.

Model	Params	Inf. (ms)	Train/ep.	Mem.
DEGRADOMAP (base)	1.43M	17.3 $\pm$ 0.6	5.19 s	75.7
DEGRADOMAP (impr.)	1.59M	—	—	—
SchNet	0.27M	4.9 $\pm$ 0.0	1.40 s	57.7
EGNN	0.44M	5.4 $\pm$ 0.3	1.70 s	60.9

DEGRADOMAP is $\sim 3.5\times$ slower per inference than SchNet but achieves +15.2% higher target-unseen AUROC. At 17.3 ms per protein, screening a library of 10,000 proteins requires ~ 3 minutes, making DEGRADOMAP practical for high-throughput virtual screening applications. Peak training memory (75.7 MB) is well within the capacity of consumer GPUs, and the entire training pipeline completes in ~ 26 hours on a single RTX 2080 Ti.

D.2 Inference Scaling by Protein Size

Table 27: Inference time by protein size for De-gradoMap.

Size	Residues	Mean res.	n	Time (ms)
Small	<200	147	8	34.9 $\pm$ 11.6
Medium	200–500	381	62	33.5 $\pm$ 12.7
Large	500–1000	771	68	42.7 $\pm$ 19.0
XLarge	>1000	1,430	33	39.9 $\pm$ 18.3

Inference time scales sub-linearly with protein size due to sparse radius graph construction: the number of edges grows proportionally to N rather than N^2 . Even the largest proteins (>1000 residues, mean 1,430) require <60 ms, with no special memory management needed.

D.3 Training Resource Requirements

Table 28: Full training pipeline resource requirements.

Resource	Requirement
GPU	NVIDIA RTX 2080 Ti (11 GB)
Total training time	~26 hours (20 epochs $\times$ 3 splits)
Per-epoch time	~5.2 seconds (full training set)
Peak GPU memory	75.7 MB
Data storage	~200 MB (structures + features)
Model checkpoint	5.5 MB

The modest computational requirements (single consumer GPU, <80 MB memory) make DEGRADOMAP accessible to academic research groups without high-performance computing infrastructure.

E Reproducibility

E.1 Complete Hyperparameter Table

Table 29: Complete hyperparameter specification for DegradomAP. The improved model column shows tuned values; the baseline column shows original values where they differ.

Category	Hyperparameter	Impr.	Base.
Arch.	GNN layers	4	4
	GNN hidden dim	128	128
	GNN output dim	64	64
	Node feat. dim	1,285	28
	Cross-attn. layers	2	2
	Attention heads	4	4
	Radius cutoff	8 Å	8 Å
	E3 embed. dim	64	64
	Ctx MLP blocks	3	3
Training	Ctx hidden dim	128	128
	Optimizer	AdamW	AdamW
	Learning rate	5×10^{-4}	10^{-3}
	β_1, β_2	0.9, 0.999	0.9, 0.999
	Weight decay	10^{-2}	10^{-2}
	LR schedule	Cos. ann.	Cos. ann.
	Batch size	8	32
	Epochs (max)	10	20
	Dropout	0.05	0.1
	Early stopping	pat.=5	–
Loss	λ_{BCE}	1.0	1.0
	λ_{Huber}	0.5	0.5
	λ_{focal}	0.3	0.3
	Focal γ	2.0	2.0
ESM-2	Model	650M	–
	Embed. dim	1,280	–

E.2 Random Seeds

All experiments use seed 42 unless otherwise noted. Cross-validation experiments (Section A.1) use seeds {42, 123, 456} per fold to assess seed sensitivity. Bootstrap confidence intervals use 1,000 iterations with sequential numpy random seeds. Multi-seed GNN baseline experiments use seeds {42, 123, 456}.

E.3 Software and Hardware

Table 30: Software and hardware specifications.

Component	Version / Specification
<i>Hardware</i>	
GPU	NVIDIA GeForce RTX 2080 Ti (11 GB)
CPU	Intel Core i9
RAM	32 GB
OS	Ubuntu Linux 5.15
<i>Software</i>	
Python	3.13
PyTorch	2.5.1+cu118
PyTorch Geometric	2.4
scikit-learn	1.3
NumPy	1.26
Pandas	2.0

E.4 Data and Code Availability

All data and code required for reproduction are publicly available:

- **PROTAC-8K dataset:** <https://zenodo.org/records/14715718>
- **AlphaFold structures:** <https://alphafold.ebi.ac.uk/> (v6 API, human proteome)
- **DepMap features:** <https://depmap.org/portal/> (24Q2 release)
- **Code and pre-trained models:** <https://github.com/anonymoussubmitter-167/DKO>

The repository includes training scripts, evaluation pipelines, baseline implementations, and pre-trained model checkpoints for all three evaluation splits. A Docker container is provided for exact environment reproduction.

E.5 GNN Baseline Implementation Details

All GNN baselines use identical graph construction (C α radius graph, 8 Å cutoff) and training procedures (AdamW, cosine annealing, 20 epochs) for fair comparison. Table ?? provides architecture specifications and complete multi-seed results.

SchNet uses 50 radial basis functions for continuous-filter convolutions. EGNN updates both node features and coordinates at each layer, with the final coordinates discarded (only scalar features used for prediction). The E(3)-equivariant variant of DEGRADOMAP replaces the invariant message passing with vector message passing and spherical harmonics edge encoding, while keeping the E3 compatibility and fusion modules identical. EGNN shows particularly high variance on target-unseen (0.453–0.641), with seed 42 achieving performance close to DEGRADOMAP (0.641 vs. 0.657) while seeds 123 and 456 perform near random. This instability suggests that EGNN’s equivariant coordinate updates may be sensitive to initialization in the low-data regime.

1973	2031
1974	2032
1975	2033
1976	2034
1977	2035
1978	2036
1979	2037
1980	2038
1981	2039
1982	2040
1983	2041
1984	2042
1985	2043
1986	2044
1987	2045
1988	2046
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2105	2163
2106	2164
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2109	2167
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2111	2169
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2114	2172
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2137	2195
2138	2196
2139	2197
2140	2198
2141	2199
2142	2200
2143	2201
2144	2202
2145	2203
2146	2204